SoccerGMR Standard Test (1036/1036) — primary

Standard-trained checkpoints | Standard Test (1036/1036) | 544 positive / 492 negative

Value above; (method - same-backbone GMR) below. G-mIoU: 0-1; others: percent / percentage points.

Backbone	Method	AUROC	mAP	mR@1	mR@5	mR+@5	N-acc τ=0.4	Rej-F1 τ=0.4	G-mIoU@1 τ=0.4	G-mIoU@3 τ=0.4	N-acc τ=0.5	Rej-F1 τ=0.5	G-mIoU@1 τ=0.5	G-mIoU@3 τ=0.5	N-acc τ=0.6	Rej-F1 τ=0.6	G-mIoU@1 τ=0.6	G-mIoU@3 τ=0.6
Moment-DETR	**GMR (baseline)**	**68.80**	**5.91**	**3.68**	**7.98**	**1.16**	**0.00**	**0.00**	**0.0577**	**0.0220**	**100.00**	**75.00**	**0.5239**	**0.4912**	**100.00**	**75.00**	**0.5239**	**0.4912**
	GMR + GEQ Frozen	68.18	5.91	3.68	7.98	1.16	100.00	75.00	0.5239	0.4912	100.00	75.00	0.5239	0.4912	100.00	75.00	0.5239	0.4912
		(-0.62)	(0.00)	(0.00)	(0.00)	(0.00)	(+100.00)	(+75.00)	(+0.4662)	(+0.4692)	(0.00)	(0.00)	(0.0000)	(0.0000)	(0.00)	(0.00)	(0.0000)	(0.0000)
	GMR + GEQ Full	68.96	6.09	3.77	8.19	0.66	99.59	74.81	0.5243	0.4901	100.00	75.00	0.5263	0.4920	100.00	75.00	0.5263	0.4920
		(+0.16)	(+0.18)	(+0.09)	(+0.21)	(-0.50)	(+99.59)	(+74.81)	(+0.4666)	(+0.4681)	(0.00)	(0.00)	(+0.0024)	(+0.0008)	(0.00)	(0.00)	(+0.0024)	(+0.0008)
	GMR + Multi	70.14	7.27	4.71	10.04	0.28	83.54	69.60	0.4421	0.4136	100.00	75.00	0.5188	0.4895	100.00	75.00	0.5188	0.4895
		(+1.34)	(+1.36)	(+1.03)	(+2.06)	(-0.88)	(+83.54)	(+69.60)	(+0.3844)	(+0.3916)	(0.00)	(0.00)	(-0.0051)	(-0.0017)	(0.00)	(0.00)	(-0.0051)	(-0.0017)
	GMR + Multi + GEQ Frozen	69.25	7.27	4.71	10.04	0.28	90.65	72.11	0.4749	0.4456	100.00	75.00	0.5188	0.4895	100.00	75.00	0.5188	0.4895
		(+0.45)	(+1.36)	(+1.03)	(+2.06)	(-0.88)	(+90.65)	(+72.11)	(+0.4172)	(+0.4236)	(0.00)	(0.00)	(-0.0051)	(-0.0017)	(0.00)	(0.00)	(-0.0051)	(-0.0017)
	GMR + Multi + GEQ Full	69.83	7.72	4.06	12.48	0.83	81.91	68.89	0.4395	0.4070	100.00	75.00	0.5237	0.4912	100.00	75.00	0.5237	0.4912
		(+1.03)	(+1.81)	(+0.38)	(+4.50)	(-0.33)	(+81.91)	(+68.89)	(+0.3818)	(+0.3850)	(0.00)	(0.00)	(-0.0002)	(0.0000)	(0.00)	(0.00)	(-0.0002)	(0.0000)
EaTR	**GMR (baseline)**	**76.56**	**16.20**	**10.64**	**21.38**	**9.59**	**48.58**	**57.11**	**0.3332**	**0.2834**	**66.67**	**65.60**	**0.4085**	**0.3600**	**87.20**	**73.46**	**0.4956**	**0.4486**
	GMR + GEQ Frozen	75.17	16.20	10.64	21.38	9.59	63.82	64.02	0.3911	0.3452	87.60	72.93	0.4933	0.4489	96.75	74.90	0.5315	0.4858
		(-1.39)	(0.00)	(0.00)	(0.00)	(0.00)	(+15.24)	(+6.91)	(+0.0579)	(+0.0618)	(+20.93)	(+7.33)	(+0.0848)	(+0.0889)	(+9.55)	(+1.44)	(+0.0359)	(+0.0372)
	GMR + GEQ Full	75.26	17.15	11.47	22.84	10.79	64.23	64.29	0.3988	0.3482	90.65	74.77	0.5111	0.4635	98.17	75.23	0.5402	0.4921
		(-1.30)	(+0.95)	(+0.83)	(+1.46)	(+1.20)	(+15.65)	(+7.18)	(+0.0656)	(+0.0648)	(+23.98)	(+9.17)	(+0.1026)	(+0.1035)	(+10.98)	(+1.77)	(+0.0446)	(+0.0435)
	GMR + Multi	77.11	17.52	11.64	23.70	11.77	44.92	55.18	0.3305	0.2702	61.59	63.79	0.3988	0.3424	75.41	69.15	0.4575	0.4012
		(+0.55)	(+1.32)	(+1.00)	(+2.32)	(+2.18)	(-3.66)	(-1.93)	(-0.0027)	(-0.0132)	(-5.08)	(-1.81)	(-0.0097)	(-0.0176)	(-11.79)	(-4.31)	(-0.0381)	(-0.0474)
	GMR + Multi + GEQ Frozen	76.69	17.52	11.64	23.70	11.77	69.51	68.13	0.4277	0.3738	81.50	71.04	0.4738	0.4243	89.63	73.01	0.5058	0.4571
		(+0.13)	(+1.32)	(+1.00)	(+2.32)	(+2.18)	(+20.93)	(+11.02)	(+0.0945)	(+0.0904)	(+14.84)	(+5.44)	(+0.0653)	(+0.0643)	(+2.44)	(-0.45)	(+0.0102)	(+0.0085)
	GMR + Multi + GEQ Full	77.27	19.88	14.05	25.31	12.24	68.90	67.60	0.4318	0.3741	81.30	71.05	0.4796	0.4260	90.65	73.48	0.5165	0.4632
		(+0.71)	(+3.68)	(+3.41)	(+3.93)	(+2.65)	(+20.33)	(+10.49)	(+0.0986)	(+0.0907)	(+14.63)	(+5.45)	(+0.0711)	(+0.0660)	(+3.46)	(+0.02)	(+0.0209)	(+0.0146)
QD-DETR	**GMR (baseline)**	**77.66**	**15.08**	**10.04**	**20.11**	**9.62**	**53.46**	**59.84**	**0.3533**	**0.3035**	**90.04**	**73.77**	**0.5058**	**0.4611**	**97.56**	**75.29**	**0.5355**	**0.4901**
	GMR + GEQ Frozen	77.42	15.08	10.04	20.11	9.62	77.24	70.11	0.4559	0.4070	92.48	74.47	0.5161	0.4711	96.95	74.88	0.5322	0.4868
		(-0.24)	(0.00)	(0.00)	(0.00)	(0.00)	(+23.78)	(+10.27)	(+0.1026)	(+0.1035)	(+2.44)	(+0.70)	(+0.0103)	(+0.0100)	(-0.61)	(-0.41)	(-0.0033)	(-0.0033)
	GMR + GEQ Full	77.66	16.54	11.48	22.09	10.14	80.49	71.35	0.4727	0.4232	93.90	74.52	0.5241	0.4769	98.17	75.23	0.5419	0.4942
		(0.00)	(+1.46)	(+1.44)	(+1.98)	(+0.52)	(+27.03)	(+11.51)	(+0.1194)	(+0.1197)	(+3.86)	(+0.75)	(+0.0183)	(+0.0158)	(+0.61)	(-0.06)	(+0.0064)	(+0.0041)
	GMR + Multi	75.56	14.67	9.95	19.50	5.60	53.86	58.37	0.3485	0.2985	91.67	74.92	0.5157	0.4690	96.54	75.28	0.5308	0.4846
		(-2.10)	(-0.41)	(-0.09)	(-0.61)	(-4.02)	(+0.41)	(-1.47)	(-0.0048)	(-0.0050)	(+1.63)	(+1.15)	(+0.0099)	(+0.0079)	(-1.02)	(-0.01)	(-0.0047)	(-0.0055)
	GMR + Multi + GEQ Frozen	74.92	14.67	9.95	19.50	5.60	74.59	68.41	0.4380	0.3917	94.51	74.82	0.5224	0.4772	97.97	74.67	0.5320	0.4886
		(-2.74)	(-0.41)	(-0.09)	(-0.61)	(-4.02)	(+21.14)	(+8.57)	(+0.0847)	(+0.0882)	(+4.47)	(+1.05)	(+0.0166)	(+0.0161)	(+0.41)	(-0.62)	(-0.0035)	(-0.0015)
	GMR + Multi + GEQ Full	74.79	13.71	9.21	18.42	6.44	94.31	74.72	0.5194	0.4756	98.98	75.15	0.5368	0.4932	99.59	74.87	0.5387	0.4951
		(-2.87)	(-1.37)	(-0.83)	(-1.69)	(-3.18)	(+40.85)	(+14.88)	(+0.1661)	(+0.1721)	(+8.94)	(+1.38)	(+0.0310)	(+0.0321)	(+2.03)	(-0.42)	(+0.0032)	(+0.0050)
CG-DETR	**GMR (baseline)**	**75.97**	**16.09**	**10.76**	**22.12**	**9.48**	**96.75**	**75.68**	**0.5382**	**0.4890**	**99.19**	**75.84**	**0.5465**	**0.4975**	**99.59**	**75.91**	**0.5484**	**0.4994**
	GMR + GEQ Frozen	76.39	16.09	10.76	22.12	9.48	75.00	69.04	0.4457	0.3969	92.48	74.23	0.5189	0.4698	99.19	75.95	0.5463	0.4976
		(+0.42)	(0.00)	(0.00)	(0.00)	(0.00)	(-21.75)	(-6.64)	(-0.0925)	(-0.0921)	(-6.71)	(-1.61)	(-0.0276)	(-0.0277)	(-0.41)	(+0.04)	(-0.0021)	(-0.0018)
	GMR + GEQ Full	74.83	16.58	11.12	22.60	10.97	75.00	68.59	0.4439	0.3933	97.36	75.55	0.5381	0.4889	99.80	75.42	0.5446	0.4978
		(-1.14)	(+0.49)	(+0.36)	(+0.48)	(+1.49)	(-21.75)	(-7.09)	(-0.0943)	(-0.0957)	(-1.83)	(-0.29)	(-0.0084)	(-0.0086)	(+0.20)	(-0.49)	(-0.0038)	(-0.0016)
	GMR + Multi	77.40	16.59	11.21	21.77	8.14	46.95	56.20	0.3369	0.2783	75.41	69.61	0.4542	0.4005	90.45	74.23	0.5167	0.4641
		(+1.43)	(+0.50)	(+0.45)	(-0.35)	(-1.34)	(-49.80)	(-19.48)	(-0.2013)	(-0.2107)	(-23.78)	(-6.23)	(-0.0923)	(-0.0970)	(-9.15)	(-1.68)	(-0.0317)	(-0.0353)
	GMR + Multi + GEQ Frozen	75.95	16.59	11.21	21.77	8.14	70.33	67.32	0.4310	0.3758	85.16	72.68	0.4938	0.4417	92.68	73.91	0.5198	0.4701
		(-0.02)	(+0.50)	(+0.45)	(-0.35)	(-1.34)	(-26.42)	(-8.36)	(-0.1072)	(-0.1132)	(-14.02)	(-3.16)	(-0.0527)	(-0.0558)	(-6.91)	(-2.00)	(-0.0286)	(-0.0293)
	GMR + Multi + GEQ Full	75.77	18.59	12.31	24.98	12.13	59.15	62.72	0.3884	0.3304	81.30	71.43	0.4784	0.4246	93.90	74.04	0.5244	0.4750
		(-0.20)	(+2.50)	(+1.55)	(+2.86)	(+2.65)	(-37.60)	(-12.96)	(-0.1498)	(-0.1586)	(-17.89)	(-4.41)	(-0.0681)	(-0.0729)	(-5.69)	(-1.87)	(-0.0240)	(-0.0244)

Green: gain Red: decrease Gray: unchanged; all deltas are relative to GMR.

N-acc = TN / negatives. τ is the existence threshold. Localization uses positive ungated candidates.

Full dataset and GEQ Full training mode are distinct. Coverage and temporal limitations: see DATA.md.

Same Standard-trained checkpoints across both Test domains. Training budgets and native seeds differ; see TRAINING.md.

SoccerGMR Standard Test (1036/1036) — localization extra

Standard-trained checkpoints | Standard Test (1036/1036) | 544 positive / 492 negative

Value above; (method - same-backbone GMR) below. G-mIoU: 0-1; others: percent / percentage points.

Backbone	Method	mR@3	mR+@1	mR+@3	mIoU@1	mIoU@3	mIoU@5	mIoU+@1	mIoU+@3	mIoU+@5
Moment-DETR	**GMR (baseline)**	**5.69**	**0.00**	**0.06**	**12.19**	**11.21**	**11.18**	**0.00**	**0.19**	**0.19**
	GMR + GEQ Frozen	5.69 (0.00)	0.00 (0.00)	0.06 (0.00)	12.19 (0.00)	11.21 (0.00)	11.18 (0.00)	0.00 (0.00)	0.19 (0.00)	0.19 (0.00)
	GMR + GEQ Full	6.38 (+0.69)	0.00 (0.00)	0.06 (0.00)	12.61 (+0.42)	11.59 (+0.38)	11.57 (+0.39)	0.00 (0.00)	0.19 (0.00)	0.12 (-0.07)
	GMR + Multi	7.04 (+1.35)	0.00 (0.00)	0.09 (+0.03)	12.53 (+0.34)	12.02 (+0.81)	11.92 (+0.74)	0.00 (0.00)	1.01 (+0.82)	0.82 (+0.63)
	GMR + Multi + GEQ Frozen	7.04 (+1.35)	0.00 (0.00)	0.09 (+0.03)	12.53 (+0.34)	12.02 (+0.81)	11.92 (+0.74)	0.00 (0.00)	1.01 (+0.82)	0.82 (+0.63)
	GMR + Multi + GEQ Full	8.74 (+3.05)	0.00 (0.00)	0.19 (+0.13)	13.63 (+1.44)	12.58 (+1.37)	12.55 (+1.37)	0.00 (0.00)	0.55 (+0.36)	0.69 (+0.50)
EaTR	**GMR (baseline)**	**18.11**	**0.00**	**7.11**	**28.33**	**25.97**	**25.77**	**0.00**	**12.94**	**12.36**
	GMR + GEQ Frozen	18.11 (0.00)	0.00 (0.00)	7.11 (0.00)	28.33 (0.00)	25.97 (0.00)	25.77 (0.00)	0.00 (0.00)	12.94 (0.00)	12.36 (0.00)
	GMR + GEQ Full	19.64 (+1.53)	0.00 (0.00)	7.79 (+0.68)	28.99 (+0.66)	27.47 (+1.50)	27.33 (+1.56)	0.00 (0.00)	13.23 (+0.29)	12.98 (+0.62)
	GMR + Multi	19.88 (+1.77)	0.00 (0.00)	6.66 (-0.45)	30.86 (+2.53)	27.97 (+2.00)	27.85 (+2.08)	0.00 (0.00)	11.07 (-1.87)	10.89 (-1.47)
	GMR + Multi + GEQ Frozen	19.88 (+1.77)	0.00 (0.00)	6.66 (-0.45)	30.86 (+2.53)	27.97 (+2.00)	27.85 (+2.08)	0.00 (0.00)	11.07 (-1.87)	10.89 (-1.47)
	GMR + Multi + GEQ Full	21.74 (+3.63)	0.00 (0.00)	7.56 (+0.45)	33.26 (+4.93)	30.13 (+4.16)	30.09 (+4.32)	0.00 (0.00)	11.31 (-1.63)	11.53 (-0.83)
QD-DETR	**GMR (baseline)**	**16.93**	**0.00**	**5.75**	**27.19**	**25.57**	**25.39**	**0.00**	**11.64**	**11.27**
	GMR + GEQ Frozen	16.93 (0.00)	0.00 (0.00)	5.75 (0.00)	27.19 (0.00)	25.57 (0.00)	25.39 (0.00)	0.00 (0.00)	11.64 (0.00)	11.27 (0.00)
	GMR + GEQ Full	18.13 (+1.20)	0.00 (0.00)	6.39 (+0.64)	29.24 (+2.05)	27.04 (+1.47)	26.92 (+1.53)	0.00 (0.00)	12.58 (+0.94)	12.43 (+1.16)
	GMR + Multi	16.16 (-0.77)	0.00 (0.00)	2.73 (-3.02)	26.26 (-0.93)	23.99 (-1.58)	23.91 (-1.48)	0.00 (0.00)	7.87 (-3.77)	7.69 (-3.58)
	GMR + Multi + GEQ Frozen	16.16 (-0.77)	0.00 (0.00)	2.73 (-3.02)	26.26 (-0.93)	23.99 (-1.58)	23.91 (-1.48)	0.00 (0.00)	7.87 (-3.77)	7.69 (-3.58)
	GMR + Multi + GEQ Full	14.40 (-2.53)	0.00 (0.00)	3.42 (-2.33)	26.61 (-0.58)	24.22 (-1.35)	24.05 (-1.34)	0.00 (0.00)	8.88 (-2.76)	8.45 (-2.82)
CG-DETR	**GMR (baseline)**	**18.09**	**0.00**	**5.38**	**29.13**	**26.71**	**26.61**	**0.00**	**11.02**	**10.95**
	GMR + GEQ Frozen	18.09 (0.00)	0.00 (0.00)	5.38 (0.00)	29.13 (0.00)	26.71 (0.00)	26.61 (0.00)	0.00 (0.00)	11.02 (0.00)	10.95 (0.00)
	GMR + GEQ Full	18.52 (+0.43)	0.00 (0.00)	6.17 (+0.79)	29.71 (+0.58)	27.51 (+0.80)	27.43 (+0.82)	0.00 (0.00)	13.00 (+1.98)	13.09 (+2.14)
	GMR + Multi	18.37 (+0.28)	0.00 (0.00)	4.52 (-0.86)	29.45 (+0.32)	27.05 (+0.34)	26.90 (+0.29)	0.00 (0.00)	9.44 (-1.58)	9.06 (-1.89)
	GMR + Multi + GEQ Frozen	18.37 (+0.28)	0.00 (0.00)	4.52 (-0.86)	29.45 (+0.32)	27.05 (+0.34)	26.90 (+0.29)	0.00 (0.00)	9.44 (-1.58)	9.06 (-1.89)
	GMR + Multi + GEQ Full	20.79 (+2.70)	0.00 (0.00)	6.45 (+1.07)	30.33 (+1.20)	28.33 (+1.62)	28.19 (+1.58)	0.00 (0.00)	12.38 (+1.36)	12.14 (+1.19)

Green: gain Red: decrease Gray: unchanged; all deltas are relative to GMR.

N-acc = TN / negatives. τ is the existence threshold. Localization uses positive ungated candidates.

Full dataset and GEQ Full training mode are distinct. Coverage and temporal limitations: see DATA.md.

Same Standard-trained checkpoints across both Test domains. Training budgets and native seeds differ; see TRAINING.md.

SoccerGMR Standard Test (1036/1036) — threshold extra

Standard-trained checkpoints | Standard Test (1036/1036) | 544 positive / 492 negative

Value above; (method - same-backbone GMR) below. G-mIoU: 0-1; others: percent / percentage points.

Backbone	Method	Acc τ=0.4	Rej-P τ=0.4	Rej-R τ=0.4	G-mIoU@5 τ=0.4	Acc τ=0.5	Rej-P τ=0.5	Rej-R τ=0.5	G-mIoU@5 τ=0.5	Acc τ=0.6	Rej-P τ=0.6	Rej-R τ=0.6	G-mIoU@5 τ=0.6
Moment-DETR	**GMR (baseline)**	**52.51**	**0.00**	**0.00**	**0.0134**	**68.34**	**60.00**	**100.00**	**0.4847**	**68.34**	**60.00**	**100.00**	**0.4847**
	GMR + GEQ Frozen	68.34 (+15.83)	60.00 (+60.00)	100.00 (+100.00)	0.4847 (+0.4713)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4847 (0.0000)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4847 (0.0000)
	GMR + GEQ Full	68.15 (+15.64)	59.90 (+59.90)	99.59 (+99.59)	0.4832 (+0.4698)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4852 (+0.0005)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4852 (+0.0005)
	GMR + Multi	65.35 (+12.84)	59.65 (+59.65)	83.54 (+83.54)	0.4070 (+0.3936)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4837 (-0.0010)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4837 (-0.0010)
	GMR + Multi + GEQ Frozen	66.70 (+14.19)	59.87 (+59.87)	90.65 (+90.65)	0.4396 (+0.4262)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4837 (-0.0010)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4837 (-0.0010)
	GMR + Multi + GEQ Full	64.86 (+12.35)	59.44 (+59.44)	81.91 (+81.91)	0.4002 (+0.3868)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4847 (0.0000)	68.34 (0.00)	60.00 (0.00)	100.00 (0.00)	0.4847 (0.0000)
EaTR	**GMR (baseline)**	**65.35**	**69.28**	**48.58**	**0.2635**	**66.80**	**64.57**	**66.67**	**0.3438**	**70.08**	**63.46**	**87.20**	**0.4358**
	GMR + GEQ Frozen	65.93 (+0.58)	64.21 (-5.07)	63.82 (+15.24)	0.3295 (+0.0660)	69.11 (+2.31)	62.46 (-2.11)	87.60 (+20.93)	0.4367 (+0.0929)	69.21 (-0.87)	61.10 (-2.36)	96.75 (+9.55)	0.4758 (+0.0400)
	GMR + GEQ Full	66.12 (+0.77)	64.36 (-4.92)	64.23 (+15.65)	0.3319 (+0.0684)	70.95 (+4.15)	63.62 (-0.95)	90.65 (+23.98)	0.4512 (+0.1074)	69.31 (-0.77)	60.98 (-2.48)	98.17 (+10.97)	0.4822 (+0.0464)
	GMR + Multi	65.35 (0.00)	71.52 (+2.24)	44.92 (-3.66)	0.2487 (-0.0148)	66.80 (0.00)	66.16 (+1.59)	61.59 (-5.08)	0.3236 (-0.0202)	68.05 (-2.03)	63.86 (+0.40)	75.41 (-11.79)	0.3850 (-0.0508)
	GMR + Multi + GEQ Frozen	69.11 (+3.76)	66.80 (-2.48)	69.51 (+20.93)	0.3574 (+0.0939)	68.44 (+1.64)	62.95 (-1.62)	81.50 (+14.83)	0.4104 (+0.0666)	68.53 (-1.55)	61.59 (-1.87)	89.63 (+2.43)	0.4454 (+0.0096)
	GMR + Multi + GEQ Full	68.63 (+3.28)	66.34 (-2.94)	68.90 (+20.32)	0.3563 (+0.0928)	68.53 (+1.73)	63.09 (-1.48)	81.30 (+14.63)	0.4109 (+0.0671)	68.92 (-1.16)	61.77 (-1.69)	90.65 (+3.45)	0.4509 (+0.0151)
QD-DETR	**GMR (baseline)**	**65.93**	**67.96**	**53.46**	**0.2848**	**69.59**	**62.48**	**90.04**	**0.4485**	**69.59**	**61.30**	**97.56**	**0.4798**
	GMR + GEQ Frozen	68.73 (+2.80)	64.19 (-3.77)	77.24 (+23.78)	0.3920 (+0.1072)	69.88 (+0.29)	62.33 (-0.15)	92.48 (+2.44)	0.4592 (+0.0107)	69.11 (-0.48)	61.00 (-0.30)	96.95 (-0.61)	0.4766 (-0.0032)
	GMR + GEQ Full	69.31 (+3.38)	64.08 (-3.88)	80.49 (+27.03)	0.4082 (+0.1234)	69.50 (-0.09)	61.76 (-0.72)	93.90 (+3.86)	0.4653 (+0.0168)	69.31 (-0.28)	60.98 (-0.32)	98.17 (+0.61)	0.4834 (+0.0036)
	GMR + Multi	63.51 (-2.42)	63.70 (-4.26)	53.86 (+0.40)	0.2826 (-0.0022)	70.85 (+1.26)	63.34 (+0.86)	91.67 (+1.63)	0.4564 (+0.0079)	69.88 (+0.29)	61.69 (+0.39)	96.54 (-1.02)	0.4747 (-0.0051)
	GMR + Multi + GEQ Frozen	67.28 (+1.35)	63.17 (-4.79)	74.59 (+21.13)	0.3778 (+0.0930)	69.79 (+0.20)	61.92 (-0.56)	94.51 (+4.47)	0.4666 (+0.0181)	68.44 (-1.15)	60.33 (-0.97)	97.97 (+0.41)	0.4799 (+0.0001)
	GMR + Multi + GEQ Full	69.69 (+3.76)	61.87 (-6.09)	94.31 (+40.85)	0.4653 (+0.1805)	68.92 (-0.67)	60.57 (-1.91)	98.98 (+8.94)	0.4841 (+0.0356)	68.24 (-1.35)	59.98 (-1.32)	99.59 (+2.03)	0.4862 (+0.0064)
CG-DETR	**GMR (baseline)**	**70.46**	**62.14**	**96.75**	**0.4784**	**69.98**	**61.38**	**99.19**	**0.4876**	**69.98**	**61.33**	**99.59**	**0.4895**
	GMR + GEQ Frozen	68.05 (-2.41)	63.95 (+1.81)	75.00 (-21.75)	0.3825 (-0.0959)	69.50 (-0.48)	61.99 (+0.61)	92.48 (-6.71)	0.4592 (-0.0284)	70.17 (+0.19)	61.54 (+0.21)	99.19 (-0.40)	0.4881 (-0.0014)
	GMR + GEQ Full	67.37 (-3.09)	63.18 (+1.04)	75.00 (-21.75)	0.3810 (-0.0974)	70.08 (+0.10)	61.73 (+0.35)	97.36 (-1.83)	0.4804 (-0.0072)	69.11 (-0.87)	60.62 (-0.71)	99.80 (+0.21)	0.4899 (+0.0004)
	GMR + Multi	65.25 (-5.21)	70.00 (+7.86)	46.95 (-49.80)	0.2571 (-0.2213)	68.73 (-1.25)	64.63 (+3.25)	75.41 (-23.78)	0.3845 (-0.1031)	70.17 (+0.19)	62.94 (+1.61)	90.45 (-9.14)	0.4507 (-0.0388)
	GMR + Multi + GEQ Frozen	67.57 (-2.89)	64.55 (+2.41)	70.33 (-26.42)	0.3598 (-0.1186)	69.59 (-0.39)	63.39 (+2.01)	85.16 (-14.03)	0.4273 (-0.0603)	68.92 (-1.06)	61.46 (+0.13)	92.68 (-6.91)	0.4586 (-0.0309)
	GMR + Multi + GEQ Full	66.60 (-3.86)	66.74 (+4.60)	59.15 (-37.60)	0.3115 (-0.1669)	69.11 (-0.87)	63.69 (+2.31)	81.30 (-17.89)	0.4097 (-0.0779)	68.73 (-1.25)	61.11 (-0.22)	93.90 (-5.69)	0.4637 (-0.0258)

Green: gain Red: decrease Gray: unchanged; all deltas are relative to GMR.

N-acc = TN / negatives. τ is the existence threshold. Localization uses positive ungated candidates.

Full dataset and GEQ Full training mode are distinct. Coverage and temporal limitations: see DATA.md.

Same Standard-trained checkpoints across both Test domains. Training budgets and native seeds differ; see TRAINING.md.

SoccerGMR Full-Decodable Test (2970/2986) — primary

Standard-trained checkpoints | Full-Decodable Test (2970/2986) | 1633 positive / 1337 negative | 16 original exclusions

Value above; (method - same-backbone GMR) below. G-mIoU: 0-1; others: percent / percentage points.

Backbone	Method	AUROC	mAP	mR@1	mR@5	mR+@5	N-acc τ=0.4	Rej-F1 τ=0.4	G-mIoU@1 τ=0.4	G-mIoU@3 τ=0.4	N-acc τ=0.5	Rej-F1 τ=0.5	G-mIoU@1 τ=0.5	G-mIoU@3 τ=0.5	N-acc τ=0.6	Rej-F1 τ=0.6	G-mIoU@1 τ=0.6	G-mIoU@3 τ=0.6
Moment-DETR	**GMR (baseline)**	**77.92**	**5.41**	**3.39**	**7.33**	**0.81**	**0.00**	**0.00**	**0.0589**	**0.0253**	**97.01**	**78.92**	**0.4885**	**0.4575**	**97.16**	**78.97**	**0.4891**	**0.4582**
	GMR + GEQ Frozen	79.03 (+1.11)	5.41 (0.00)	3.39 (0.00)	7.33 (0.00)	0.81 (0.00)	96.41 (+96.41)	78.74 (+78.74)	0.4858 (+0.4269)	0.4548 (+0.4295)	97.16 (+0.15)	78.97 (+0.05)	0.4891 (+0.0006)	0.4582 (+0.0007)	97.16 (0.00)	78.82 (-0.15)	0.4888 (-0.0003)	0.4581 (-0.0001)
	GMR + GEQ Full	78.62 (+0.70)	5.28 (-0.13)	3.31 (-0.08)	7.33 (0.00)	0.65 (-0.16)	91.92 (+91.92)	77.15 (+77.15)	0.4648 (+0.4059)	0.4340 (+0.4087)	96.63 (-0.37)	78.80 (-0.12)	0.4857 (-0.0028)	0.4550 (-0.0025)	97.08 (-0.07)	78.93 (-0.04)	0.4877 (-0.0014)	0.4570 (-0.0012)
	GMR + Multi	79.25 (+1.33)	6.06 (+0.65)	3.86 (+0.47)	8.63 (+1.30)	0.49 (-0.32)	83.62 (+83.62)	74.26 (+74.26)	0.4237 (+0.3648)	0.3964 (+0.3711)	97.16 (+0.15)	78.89 (-0.03)	0.4839 (-0.0046)	0.4563 (-0.0012)	97.16 (0.00)	78.89 (-0.08)	0.4839 (-0.0052)	0.4563 (-0.0019)
	GMR + Multi + GEQ Frozen	78.81 (+0.89)	6.06 (+0.65)	3.86 (+0.47)	8.63 (+1.30)	0.49 (-0.32)	88.93 (+88.93)	76.12 (+76.12)	0.4471 (+0.3882)	0.4196 (+0.3943)	97.16 (+0.15)	78.92 (0.00)	0.4839 (-0.0046)	0.4563 (-0.0012)	97.16 (0.00)	78.92 (-0.05)	0.4839 (-0.0052)	0.4563 (-0.0019)
	GMR + Multi + GEQ Full	78.97 (+1.05)	6.39 (+0.98)	3.38 (-0.01)	10.09 (+2.76)	0.77 (-0.04)	81.82 (+81.82)	73.40 (+73.40)	0.4196 (+0.3607)	0.3902 (+0.3649)	97.08 (+0.07)	78.86 (-0.06)	0.4870 (-0.0015)	0.4578 (+0.0003)	97.16 (0.00)	78.87 (-0.10)	0.4874 (-0.0017)	0.4581 (-0.0001)
EaTR	**GMR (baseline)**	**81.59**	**15.24**	**10.11**	**20.37**	**6.99**	**47.87**	**57.25**	**0.3208**	**0.2701**	**66.12**	**67.38**	**0.3961**	**0.3467**	**85.42**	**75.06**	**0.4739**	**0.4264**
	GMR + GEQ Frozen	82.57 (+0.98)	15.24 (0.00)	10.11 (0.00)	20.37 (0.00)	6.99 (0.00)	57.07 (+9.20)	63.40 (+6.15)	0.3577 (+0.0369)	0.3090 (+0.0389)	77.56 (+11.44)	73.13 (+5.75)	0.4421 (+0.0460)	0.3959 (+0.0492)	88.93 (+3.52)	76.71 (+1.65)	0.4870 (+0.0131)	0.4420 (+0.0156)
	GMR + GEQ Full	82.34 (+0.75)	15.27 (+0.03)	10.42 (+0.31)	20.13 (-0.24)	7.48 (+0.49)	55.80 (+7.93)	62.45 (+5.20)	0.3527 (+0.0319)	0.3021 (+0.0320)	80.25 (+14.14)	74.05 (+6.67)	0.4517 (+0.0556)	0.4047 (+0.0580)	91.55 (+6.13)	77.42 (+2.36)	0.4975 (+0.0236)	0.4512 (+0.0248)
	GMR + Multi	82.43 (+0.84)	16.12 (+0.88)	10.77 (+0.66)	21.85 (+1.48)	7.49 (+0.50)	45.33 (-2.54)	56.53 (-0.72)	0.3171 (-0.0037)	0.2610 (-0.0091)	62.30 (-3.81)	66.64 (-0.74)	0.3873 (-0.0088)	0.3331 (-0.0136)	75.02 (-10.40)	71.64 (-3.42)	0.4376 (-0.0363)	0.3855 (-0.0409)
	GMR + Multi + GEQ Frozen	83.46 (+1.87)	16.12 (+0.88)	10.77 (+0.66)	21.85 (+1.48)	7.49 (+0.50)	66.34 (+18.47)	69.13 (+11.88)	0.4007 (+0.0799)	0.3492 (+0.0791)	79.21 (+13.09)	74.21 (+6.83)	0.4515 (+0.0554)	0.4030 (+0.0563)	88.11 (+2.69)	76.97 (+1.91)	0.4868 (+0.0129)	0.4395 (+0.0131)
	GMR + Multi + GEQ Full	83.43 (+1.84)	16.81 (+1.57)	11.65 (+1.54)	21.79 (+1.42)	7.47 (+0.48)	67.17 (+19.30)	69.40 (+12.15)	0.4065 (+0.0857)	0.3537 (+0.0836)	80.33 (+14.21)	74.82 (+7.44)	0.4596 (+0.0635)	0.4092 (+0.0625)	89.01 (+3.59)	77.12 (+2.06)	0.4930 (+0.0191)	0.4438 (+0.0174)
QD-DETR	**GMR (baseline)**	**83.62**	**12.31**	**8.81**	**16.05**	**4.11**	**52.13**	**61.27**	**0.3322**	**0.2817**	**87.81**	**77.24**	**0.4793**	**0.4326**	**95.06**	**79.04**	**0.5076**	**0.4611**
	GMR + GEQ Frozen	83.14 (-0.48)	12.31 (0.00)	8.81 (0.00)	16.05 (0.00)	4.11 (0.00)	72.10 (+19.97)	71.46 (+10.19)	0.4158 (+0.0836)	0.3662 (+0.0845)	87.58 (-0.22)	76.86 (-0.38)	0.4767 (-0.0026)	0.4301 (-0.0025)	93.27 (-1.80)	78.33 (-0.71)	0.4999 (-0.0077)	0.4534 (-0.0077)
	GMR + GEQ Full	83.29 (-0.33)	13.50 (+1.19)	9.65 (+0.84)	17.70 (+1.65)	4.52 (+0.41)	77.49 (+25.36)	73.92 (+12.65)	0.4404 (+0.1082)	0.3900 (+0.1083)	90.88 (+3.07)	77.76 (+0.52)	0.4923 (+0.0130)	0.4448 (+0.0122)	94.84 (-0.22)	78.86 (-0.18)	0.5080 (+0.0004)	0.4607 (-0.0004)
	GMR + Multi	82.03 (-1.59)	12.04 (-0.27)	8.43 (-0.38)	15.54 (-0.51)	3.24 (-0.87)	54.53 (+2.39)	61.65 (+0.38)	0.3327 (+0.0005)	0.2857 (+0.0040)	87.28 (-0.52)	76.70 (-0.54)	0.4708 (-0.0085)	0.4267 (-0.0059)	93.12 (-1.94)	78.20 (-0.84)	0.4926 (-0.0150)	0.4496 (-0.0115)
	GMR + Multi + GEQ Frozen	82.44 (-1.18)	12.04 (-0.27)	8.43 (-0.38)	15.54 (-0.51)	3.24 (-0.87)	68.59 (+16.45)	69.18 (+7.91)	0.3903 (+0.0581)	0.3460 (+0.0643)	88.48 (+0.67)	77.14 (-0.10)	0.4735 (-0.0058)	0.4303 (-0.0023)	93.87 (-1.20)	78.19 (-0.85)	0.4934 (-0.0142)	0.4518 (-0.0093)
	GMR + Multi + GEQ Full	82.42 (-1.20)	11.85 (-0.46)	8.34 (-0.47)	15.26 (-0.79)	3.41 (-0.70)	88.71 (+36.57)	77.24 (+15.97)	0.4765 (+0.1443)	0.4334 (+0.1517)	95.44 (+7.63)	78.81 (+1.57)	0.5026 (+0.0233)	0.4606 (+0.0280)	96.86 (+1.80)	78.89 (-0.15)	0.5075 (-0.0001)	0.4660 (+0.0049)
CG-DETR	**GMR (baseline)**	**82.34**	**13.58**	**9.93**	**17.97**	**5.59**	**94.39**	**78.43**	**0.5068**	**0.4586**	**96.56**	**78.62**	**0.5143**	**0.4660**	**96.78**	**78.52**	**0.5150**	**0.4667**
	GMR + GEQ Frozen	82.90 (+0.56)	13.58 (0.00)	9.93 (0.00)	17.97 (0.00)	5.59 (0.00)	76.74 (-17.65)	73.29 (-5.14)	0.4360 (-0.0708)	0.3869 (-0.0717)	91.92 (-4.64)	78.28 (-0.34)	0.4982 (-0.0161)	0.4496 (-0.0164)	96.19 (-0.60)	79.19 (+0.67)	0.5141 (-0.0009)	0.4658 (-0.0009)
	GMR + GEQ Full	81.82 (-0.52)	14.01 (+0.43)	10.23 (+0.30)	18.77 (+0.80)	5.31 (-0.28)	75.39 (-19.00)	72.03 (-6.40)	0.4316 (-0.0752)	0.3797 (-0.0789)	94.32 (-2.24)	78.94 (+0.32)	0.5092 (-0.0051)	0.4593 (-0.0067)	96.86 (+0.07)	79.11 (+0.59)	0.5176 (+0.0026)	0.4690 (+0.0023)
	GMR + Multi	81.02 (-1.32)	15.48 (+1.90)	10.52 (+0.59)	20.42 (+2.45)	6.12 (+0.53)	51.16 (-43.23)	59.50 (-18.93)	0.3398 (-0.1670)	0.2823 (-0.1763)	76.66 (-19.90)	71.45 (-7.17)	0.4423 (-0.0720)	0.3887 (-0.0773)	88.86 (-7.93)	74.76 (-3.76)	0.4893 (-0.0257)	0.4378 (-0.0289)
	GMR + Multi + GEQ Frozen	83.01 (+0.67)	15.48 (+1.90)	10.52 (+0.59)	20.42 (+2.45)	6.12 (+0.53)	71.28 (-23.11)	70.59 (-7.84)	0.4200 (-0.0868)	0.3669 (-0.0917)	84.29 (-12.27)	75.79 (-2.83)	0.4727 (-0.0416)	0.4222 (-0.0438)	91.32 (-5.46)	77.33 (-1.19)	0.4986 (-0.0164)	0.4497 (-0.0170)
	GMR + Multi + GEQ Full	82.81 (+0.47)	15.93 (+2.35)	10.75 (+0.82)	21.44 (+3.47)	7.62 (+2.03)	56.10 (-38.29)	63.34 (-15.09)	0.3593 (-0.1475)	0.3041 (-0.1545)	77.94 (-18.62)	73.56 (-5.06)	0.4455 (-0.0688)	0.3947 (-0.0713)	89.75 (-7.03)	77.20 (-1.32)	0.4929 (-0.0221)	0.4435 (-0.0232)

Green: gain Red: decrease Gray: unchanged; all deltas are relative to GMR.

N-acc = TN / negatives. τ is the existence threshold. Localization uses positive ungated candidates.

Full dataset and GEQ Full training mode are distinct. Coverage and temporal limitations: see DATA.md.

Same Standard-trained checkpoints across both Test domains. Training budgets and native seeds differ; see TRAINING.md.

SoccerGMR Full-Decodable Test (2970/2986) — localization extra

Standard-trained checkpoints | Full-Decodable Test (2970/2986) | 1633 positive / 1337 negative | 16 original exclusions

Value above; (method - same-backbone GMR) below. G-mIoU: 0-1; others: percent / percentage points.

Backbone	Method	mR@3	mR+@1	mR+@3	mIoU@1	mIoU@3	mIoU@5	mIoU+@1	mIoU+@3	mIoU+@5
Moment-DETR	**GMR (baseline)**	**5.56**	**0.00**	**0.26**	**12.33**	**11.42**	**11.37**	**0.00**	**0.30**	**0.28**
	GMR + GEQ Frozen	5.56 (0.00)	0.00 (0.00)	0.26 (0.00)	12.33 (0.00)	11.42 (0.00)	11.37 (0.00)	0.00 (0.00)	0.30 (0.00)	0.28 (0.00)
	GMR + GEQ Full	5.77 (+0.21)	0.00 (0.00)	0.10 (-0.16)	12.12 (-0.21)	11.02 (-0.40)	11.00 (-0.37)	0.00 (0.00)	0.31 (+0.01)	0.28 (0.00)
	GMR + Multi	6.32 (+0.76)	0.00 (0.00)	0.26 (0.00)	11.77 (-0.56)	11.10 (-0.32)	11.04 (-0.33)	0.00 (0.00)	0.68 (+0.38)	0.61 (+0.33)
	GMR + Multi + GEQ Frozen	6.32 (+0.76)	0.00 (0.00)	0.26 (0.00)	11.77 (-0.56)	11.10 (-0.32)	11.04 (-0.33)	0.00 (0.00)	0.68 (+0.38)	0.61 (+0.33)
	GMR + Multi + GEQ Full	7.20 (+1.64)	0.00 (0.00)	0.12 (-0.14)	12.57 (+0.24)	11.63 (+0.21)	11.59 (+0.22)	0.00 (0.00)	0.53 (+0.23)	0.55 (+0.27)
EaTR	**GMR (baseline)**	**17.04**	**0.00**	**4.44**	**27.60**	**25.15**	**25.03**	**0.00**	**9.29**	**9.05**
	GMR + GEQ Frozen	17.04 (0.00)	0.00 (0.00)	4.44 (0.00)	27.60 (0.00)	25.15 (0.00)	25.03 (0.00)	0.00 (0.00)	9.29 (0.00)	9.05 (0.00)
	GMR + GEQ Full	16.88 (-0.16)	0.00 (0.00)	3.99 (-0.45)	27.73 (+0.13)	25.27 (+0.12)	25.15 (+0.12)	0.00 (0.00)	8.46 (-0.83)	8.27 (-0.78)
	GMR + Multi	18.43 (+1.39)	0.00 (0.00)	4.63 (+0.19)	28.75 (+1.15)	25.90 (+0.75)	25.80 (+0.77)	0.00 (0.00)	7.94 (-1.35)	7.82 (-1.23)
	GMR + Multi + GEQ Frozen	18.43 (+1.39)	0.00 (0.00)	4.63 (+0.19)	28.75 (+1.15)	25.90 (+0.75)	25.80 (+0.77)	0.00 (0.00)	7.94 (-1.35)	7.82 (-1.23)
	GMR + Multi + GEQ Full	18.68 (+1.64)	0.00 (0.00)	4.35 (-0.09)	29.22 (+1.62)	26.47 (+1.32)	26.40 (+1.37)	0.00 (0.00)	7.59 (-1.70)	7.59 (-1.46)
QD-DETR	**GMR (baseline)**	**13.48**	**0.00**	**2.49**	**24.29**	**22.59**	**22.50**	**0.00**	**5.16**	**5.04**
	GMR + GEQ Frozen	13.48 (0.00)	0.00 (0.00)	2.49 (0.00)	24.29 (0.00)	22.59 (0.00)	22.50 (0.00)	0.00 (0.00)	5.16 (0.00)	5.04 (0.00)
	GMR + GEQ Full	15.03 (+1.55)	0.00 (0.00)	2.60 (+0.11)	25.54 (+1.25)	23.30 (+0.71)	23.23 (+0.73)	0.00 (0.00)	5.36 (+0.20)	5.31 (+0.27)
	GMR + Multi	12.90 (-0.58)	0.00 (0.00)	1.37 (-1.12)	22.90 (-1.39)	20.79 (-1.80)	20.70 (-1.80)	0.00 (0.00)	4.36 (-0.80)	4.17 (-0.87)
	GMR + Multi + GEQ Frozen	12.90 (-0.58)	0.00 (0.00)	1.37 (-1.12)	22.90 (-1.39)	20.79 (-1.80)	20.70 (-1.80)	0.00 (0.00)	4.36 (-0.80)	4.17 (-0.87)
	GMR + Multi + GEQ Full	12.69 (-0.79)	0.00 (0.00)	1.92 (-0.57)	23.99 (-0.30)	21.72 (-0.87)	21.61 (-0.89)	0.00 (0.00)	4.77 (-0.39)	4.55 (-0.49)
CG-DETR	**GMR (baseline)**	**15.19**	**0.00**	**3.38**	**26.31**	**23.63**	**23.56**	**0.00**	**5.76**	**5.68**
	GMR + GEQ Frozen	15.19 (0.00)	0.00 (0.00)	3.38 (0.00)	26.31 (0.00)	23.63 (0.00)	23.56 (0.00)	0.00 (0.00)	5.76 (0.00)	5.68 (0.00)
	GMR + GEQ Full	15.62 (+0.43)	0.00 (0.00)	3.12 (-0.26)	26.98 (+0.67)	23.98 (+0.35)	23.90 (+0.34)	0.00 (0.00)	5.60 (-0.16)	5.57 (-0.11)
	GMR + Multi	17.24 (+2.05)	0.00 (0.00)	3.46 (+0.08)	27.77 (+1.46)	24.91 (+1.28)	24.81 (+1.25)	0.00 (0.00)	5.93 (+0.17)	5.80 (+0.12)
	GMR + Multi + GEQ Frozen	17.24 (+2.05)	0.00 (0.00)	3.46 (+0.08)	27.77 (+1.46)	24.91 (+1.28)	24.81 (+1.25)	0.00 (0.00)	5.93 (+0.17)	5.80 (+0.12)
	GMR + Multi + GEQ Full	18.02 (+2.83)	0.00 (0.00)	4.06 (+0.68)	27.81 (+1.50)	25.41 (+1.78)	25.32 (+1.76)	0.00 (0.00)	7.01 (+1.25)	6.91 (+1.23)

Green: gain Red: decrease Gray: unchanged; all deltas are relative to GMR.

N-acc = TN / negatives. τ is the existence threshold. Localization uses positive ungated candidates.

Full dataset and GEQ Full training mode are distinct. Coverage and temporal limitations: see DATA.md.

Same Standard-trained checkpoints across both Test domains. Training budgets and native seeds differ; see TRAINING.md.

SoccerGMR Full-Decodable Test (2970/2986) — threshold extra

Standard-trained checkpoints | Full-Decodable Test (2970/2986) | 1633 positive / 1337 negative | 16 original exclusions

Value above; (method - same-backbone GMR) below. G-mIoU: 0-1; others: percent / percentage points.

Backbone	Method	Acc τ=0.4	Rej-P τ=0.4	Rej-R τ=0.4	G-mIoU@5 τ=0.4	Acc τ=0.5	Rej-P τ=0.5	Rej-R τ=0.5	G-mIoU@5 τ=0.5	Acc τ=0.6	Rej-P τ=0.6	Rej-R τ=0.6	G-mIoU@5 τ=0.6
Moment-DETR	**GMR (baseline)**	**54.98**	**0.00**	**0.00**	**0.0153**	**76.67**	**66.51**	**97.01**	**0.4493**	**76.70**	**66.51**	**97.16**	**0.4500**
	GMR + GEQ Frozen	76.57 (+21.59)	66.55 (+66.55)	96.41 (+96.41)	0.4466 (+0.4313)	76.70 (+0.03)	66.51 (0.00)	97.16 (+0.15)	0.4500 (+0.0007)	76.50 (-0.20)	66.31 (-0.20)	97.16 (0.00)	0.4499 (-0.0001)
	GMR + GEQ Full	75.49 (+20.51)	66.47 (+66.47)	91.92 (+91.92)	0.4260 (+0.4107)	76.60 (-0.07)	66.53 (+0.02)	96.63 (-0.38)	0.4471 (-0.0022)	76.67 (-0.03)	66.50 (-0.01)	97.08 (-0.08)	0.4491 (-0.0009)
	GMR + Multi	73.91 (+18.93)	66.79 (+66.79)	83.62 (+83.62)	0.3886 (+0.3733)	76.60 (-0.07)	66.41 (-0.10)	97.16 (+0.15)	0.4488 (-0.0005)	76.60 (-0.10)	66.41 (-0.10)	97.16 (0.00)	0.4488 (-0.0012)
	GMR + Multi + GEQ Frozen	74.88 (+19.90)	66.54 (+66.54)	88.93 (+88.93)	0.4120 (+0.3967)	76.63 (-0.04)	66.45 (-0.06)	97.16 (+0.15)	0.4488 (-0.0005)	76.63 (-0.07)	66.45 (-0.06)	97.16 (0.00)	0.4488 (-0.0012)
	GMR + Multi + GEQ Full	73.30 (+18.32)	66.55 (+66.55)	81.82 (+81.82)	0.3817 (+0.3664)	76.57 (-0.10)	66.39 (-0.12)	97.08 (+0.07)	0.4496 (+0.0003)	76.57 (-0.13)	66.38 (-0.13)	97.16 (0.00)	0.4499 (-0.0001)
EaTR	**GMR (baseline)**	**67.81**	**71.19**	**47.87**	**0.2490**	**71.18**	**68.69**	**66.12**	**0.3278**	**74.44**	**66.94**	**85.42**	**0.4104**
	GMR + GEQ Frozen	70.34 (+2.53)	71.31 (+0.12)	57.07 (+9.20)	0.2890 (+0.0400)	74.34 (+3.16)	69.18 (+0.49)	77.56 (+11.44)	0.3779 (+0.0501)	75.69 (+1.25)	67.44 (+0.50)	88.93 (+3.51)	0.4259 (+0.0155)
	GMR + GEQ Full	69.80 (+1.99)	70.91 (-0.28)	55.80 (+7.93)	0.2825 (+0.0335)	74.68 (+3.50)	68.74 (+0.05)	80.25 (+14.13)	0.3881 (+0.0603)	75.96 (+1.52)	67.07 (+0.13)	91.55 (+6.13)	0.4361 (+0.0257)
	GMR + Multi	68.62 (+0.81)	75.09 (+3.90)	45.33 (-2.54)	0.2390 (-0.0100)	71.92 (+0.74)	71.63 (+2.94)	62.30 (-3.82)	0.3128 (-0.0150)	73.27 (-1.17)	68.56 (+1.62)	75.02 (-10.40)	0.3671 (-0.0433)
	GMR + Multi + GEQ Frozen	73.33 (+5.52)	72.17 (+0.98)	66.34 (+18.47)	0.3297 (+0.0807)	75.22 (+4.04)	69.81 (+1.12)	79.21 (+13.09)	0.3851 (+0.0573)	76.26 (+1.82)	68.33 (+1.39)	88.11 (+2.69)	0.4229 (+0.0125)
	GMR + Multi + GEQ Full	73.33 (+5.52)	71.78 (+0.59)	67.17 (+19.30)	0.3338 (+0.0848)	75.66 (+4.48)	70.01 (+1.32)	80.33 (+14.21)	0.3907 (+0.0629)	76.23 (+1.79)	68.04 (+1.10)	89.01 (+3.59)	0.4271 (+0.0167)
QD-DETR	**GMR (baseline)**	**70.34**	**74.31**	**52.13**	**0.2636**	**76.70**	**68.94**	**87.81**	**0.4182**	**77.31**	**67.64**	**95.06**	**0.4482**
	GMR + GEQ Frozen	74.07 (+3.73)	70.83 (-3.48)	72.10 (+19.97)	0.3501 (+0.0865)	76.26 (-0.44)	68.48 (-0.46)	87.58 (-0.23)	0.4162 (-0.0020)	76.77 (-0.54)	67.51 (-0.13)	93.27 (-1.79)	0.4404 (-0.0078)
	GMR + GEQ Full	75.39 (+5.05)	70.67 (-3.64)	77.49 (+25.36)	0.3743 (+0.1107)	76.60 (-0.10)	67.95 (-0.99)	90.88 (+3.07)	0.4311 (+0.0129)	77.10 (-0.21)	67.48 (-0.16)	94.84 (-0.22)	0.4476 (-0.0006)
	GMR + Multi	69.46 (-0.88)	70.91 (-3.40)	54.53 (+2.40)	0.2702 (+0.0066)	76.13 (-0.57)	68.41 (-0.53)	87.28 (-0.53)	0.4137 (-0.0045)	76.63 (-0.68)	67.41 (-0.23)	93.12 (-1.94)	0.4378 (-0.0104)
	GMR + Multi + GEQ Frozen	72.49 (+2.15)	69.79 (-4.52)	68.59 (+16.46)	0.3316 (+0.0680)	76.40 (-0.30)	68.38 (-0.56)	88.48 (+0.67)	0.4179 (-0.0003)	76.43 (-0.88)	67.00 (-0.64)	93.87 (-1.19)	0.4405 (-0.0077)
	GMR + Multi + GEQ Full	76.46 (+6.12)	68.40 (-5.91)	88.71 (+36.58)	0.4203 (+0.1567)	76.90 (+0.20)	67.12 (-1.82)	95.44 (+7.63)	0.4485 (+0.0303)	76.67 (-0.64)	66.55 (-1.09)	96.86 (+1.80)	0.4543 (+0.0061)
CG-DETR	**GMR (baseline)**	**76.63**	**67.09**	**94.39**	**0.4461**	**76.36**	**66.31**	**96.56**	**0.4543**	**76.16**	**66.05**	**96.78**	**0.4551**
	GMR + GEQ Frozen	74.81 (-1.82)	70.13 (+3.04)	76.74 (-17.65)	0.3716 (-0.0745)	77.04 (+0.68)	68.16 (+1.85)	91.92 (-4.64)	0.4365 (-0.0178)	77.24 (+1.08)	67.29 (+1.24)	96.19 (-0.59)	0.4537 (-0.0014)
	GMR + GEQ Full	73.64 (-2.99)	68.95 (+1.86)	75.39 (-19.00)	0.3658 (-0.0803)	77.34 (+0.98)	67.87 (+1.56)	94.32 (-2.24)	0.4474 (-0.0069)	76.97 (+0.81)	66.86 (+0.81)	96.86 (+0.08)	0.4576 (+0.0025)
	GMR + Multi	68.65 (-7.98)	71.10 (+4.01)	51.16 (-43.23)	0.2622 (-0.1839)	72.42 (-3.94)	66.91 (+0.60)	76.66 (-19.90)	0.3718 (-0.0825)	73.00 (-3.16)	64.53 (-1.52)	88.86 (-7.92)	0.4230 (-0.0321)
	GMR + Multi + GEQ Frozen	73.27 (-3.36)	69.92 (+2.83)	71.28 (-23.11)	0.3490 (-0.0971)	75.76 (-0.60)	68.85 (+2.54)	84.29 (-12.27)	0.4056 (-0.0487)	75.89 (-0.27)	67.05 (+1.00)	91.32 (-5.46)	0.4348 (-0.0203)
	GMR + Multi + GEQ Full	70.77 (-5.86)	72.74 (+5.65)	56.10 (-38.29)	0.2840 (-0.1621)	74.78 (-1.58)	69.65 (+3.34)	77.94 (-18.62)	0.3776 (-0.0767)	76.13 (-0.03)	67.72 (+1.67)	89.75 (-7.03)	0.4281 (-0.0270)

Green: gain Red: decrease Gray: unchanged; all deltas are relative to GMR.

N-acc = TN / negatives. τ is the existence threshold. Localization uses positive ungated candidates.

Full dataset and GEQ Full training mode are distinct. Coverage and temporal limitations: see DATA.md.

Same Standard-trained checkpoints across both Test domains. Training budgets and native seeds differ; see TRAINING.md.